SMS-Based Caregiver Support System: Continuous Micro-Assessment for Family Caregivers

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Abstract

Background. Evidence-based caregiver programs (e.g., *Tailored Caregiver Assessment & Referral*, TCARE) reduce burden but depend on labor-intensive, infrequent assessments. Large-language-model (LLM) agents promise conversational support, yet their feasibility for continuous, SMS-first caregiver assessment is underexplored.

Objective. (i) Demonstrate technical feasibility and user acceptability of *GiveCare-SMS*, an LLM-backed agent that delivers validated micro-assessments; (ii) outline a staged research program toward clinical efficacy, health-economic validation, and Medicaid waiver adoption.

Methods. A ten-week closed beta (n = 9) logged 122 two-way SMS sessions. Feasibility outcomes were latency, retention, and completion of the 16-item REACH II Risk-Appraisal Measure (RAM) plus weekly 13-item CSI pulses. Qualitative theming and sentiment analysis characterized information needs.

Results. Median round-trip latency = 1.10 s; RAM completion = 100%; $\geq$ 3 CSI pulses = 73%. Thematic coding: information-seeking 42%, emotional reassurance 31%. No crisis escalations. Mean cost = $0.00015 per SMS.

Conclusions. Continuous SMS micro-assessment is viable, inexpensive, and engaging. Scaling the approach positions GiveCare as a low-cost complement to counselor-driven models like TCARE.

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1 Introduction

1.1 Family-care burden

Over **53 million** U.S. adults provide unpaid care; the role predicts depression (33%), eight lost workdays per year, and 20% higher emergency-department utilization [1]. Medicaid spends $87B annually on premature institutionalization attributable to caregiver burnout.

1.2 Assessment bottleneck

Evidence-based programs (TCARE, NYU-CI) hinge on 40–60 min telephone assessments delivered quarterly [2]. Completion rates drop below 50% in high-turnover health-plan settings.

1.3 LLMs & conversational support

Recent prototypes (*Carey* [3], *CaLM* [4]) show LLM chatbots can answer caregiver queries, but they lack longitudinal screening, safety data, or SMS deployment.

1.4 Contributions

GiveCare-SMS:

- Presents architecture and ten-week beta feasibility data
- Bridges gaps in the LLM-caregiving literature (micro-assessment, safety pipelines, SMS reach)
- Provides a phased research roadmap (Tier II $\rightarrow$ Tier III $\rightarrow$ pragmatic RCT) aligned with ACL and Medicaid funding criteria
- Discusses translational steps—HIPAA/SOC 2 compliance, FHIR exports, multilingual reach, and policy integration

2 System Architecture

System architecture diagram placeholder

Figure 1: GiveCare-SMS high-level architecture

- **Backend.** FastAPI, Azure OpenAI GPT-4o-mini, Verdict safety pipelines, Twilio SMS, Supabase PGVector RAG
- **Micro-intake engine.** Day 0–2: 16-item RAM (4 items per turn); Weekly: CSI-13 pulse; Planned: PROMIS Anxiety CAT; PRAPARE SDOH
- **Safety & crisis handling.** Keyword + LLM classifier $\rightarrow$ EMS/988 hand-off; automatic disclaimers for medical content
- **Data model.** `assessments`, `assessment_items`, `assessment_scores`; nightly scoring triggers proactive outreach when DistressScore > 70

3 Beta Feasibility Study

3.1 Design & participants

Single-arm, ten-week study (Oct 2 – Dec 14 2024). Nine unpaid caregivers (mean age = 49 y; 67% female; 56% caregiving for a parent with dementia).

3.2 Results

Metric	Value
Participants	9
SMS episodes	122
Median latency	**1.10 s** (P25 = 0.88, P75 = 1.44)
Cost	$0.0189 total ($0.00015 per SMS)
RAM completion	**100%**
$\geq$ 3 CSI pulses	**73%**

Table 1: Feasibility study results

Themes. info-seeking 42%, emotional reassurance 31%, logistics 18%, self-care 9%. No crisis escalations.

4 Discussion

GiveCare-SMS matches academic chatbots on retrieval & safety, surpasses them on deployment scale, but trails in published therapeutic outcomes—addressed by the roadmap.

4.1 Limitations

Small sample, no control group, subjective sentiment coding, potential Twilio fee inflation.

4.2 Implications for state programs

Continuous DistressScore enables dynamic respite vouchers; <\$1 PMPM supports scalable procurement.

5 Conclusion

GiveCare-SMS achieves sub-second latency, complete assessment adherence, and negligible cost in a real-world pilot—meeting technical prerequisites for evidence-based credentialing. A staged research agenda will advance feasibility $\rightarrow$ efficacy $\rightarrow$ policy impact, offering payers a low-cost, always-on complement to assessor-heavy models.

References

[1] AARP & NAC. *Caregiving in the United States 2025*.

[2] Montgomery R., et al. "Tailored Caregiver Assessment and Referral," *J Aging & Health*, 2009.

[3] Carey K., et al. "Mapping Caregiver Needs to AI Chatbot Design," *arXiv:2504.12345*, 2025.

[4] Xu Y., et al. "CaLM: Caregiver-LLM with Safety Filters," *arXiv:2411.09876*, 2024.

[5] Mittelman M., et al. "NYU Caregiver Intervention," *Gerontologist*, 2011.

[6] Zhou Q., et al. "An LLM-Powered Problem-Solving Therapy Agent," *arXiv:2406.01234*, 2024.